Supplementary Material: What Spatial Fine-Tuning Actually Changes

Anonymous WACV Datasets Track submission

Paper ID 3083

1. Protocol Freeze and Governance

The seed campaign was governed by a frozen specification (configs/seed_campaign/SEED_CAMPAIGN.json) and a decision log (SPATIAL_REASONING_DECISION_LOG.md). Key entries: the battery drift correction (2026-08-11) reinstate the committed legacy Tier-A/B/C drivers; the training deadlock incident (2026-08-12) and its non-recurrence on fresh boxes; the campaign completion entry (2026-08-14) with framing decisions; and the numerical-audit/claim-hierarchy/literature-audit entries (2026-08-14).

2. Frozen Metric Definitions

From the committed analyzer (analyze_tier_c.py), read as the authoritative definitions:

- $A_{\text{transform}}(m) = P(\text{transformed prediction} = \text{expected transformed label})$: transformed-answer accuracy.
- $C_{\text{pair}}(m) = P(\text{pair consistency})$: the model's two answers on the same example obey the linked-answer law. hflip_flip: $P(\text{mirrored} \neq \text{normal})$ (answer-update rate); hflip_invariant: $P(\text{mirrored} = \text{normal})$ (response-stability rate); relcomp/facingcomp: flip-law compliance. Invalid outputs count as non-consistent.
- both_correct$(m) = P(\text{normal-correct} \wedge \text{transformed obeys the law})$: joint correctness.
- $G = A_{\text{normal}} - A_{\text{shuffle}}$ (correct-image dependence); $\Delta G = G_{\text{tuned}} - G_{\text{zero-shot}}$.

3. Independent Numerical Audit (verbatim summary)

scripts/audit_paper2_numbers_independent.py recomputes every headline quantity from raw prediction CSVs/JSONL and frozen manifests without importing any analyzer module. Hard-fail rules: deterministic proportion discrepancy $> 10^{-6}$; row-count mismatch; ID mismatch; terminology mismatch. Verdict: PASS (0 issues, 0 failed claims). Committed seed-0 reproduction: 7B normal 0.82414579 (diff 0.0); 2B normal 0.76492027 (diff 0.0). Claim-level audits: ΔA positive all fresh seeds both backbones; ΔG positive all fresh seeds both backbones; fresh-seed C_{pair} within 0.05 of legacy General; all fresh 2B hflip $C_{\text{pair}} >$ 2B zero-shot; Qwen3-VL-8B scoped. Hostile tests (single-prediction flips) produced FAIL as expected and PASS after restore.

4. Canonical Subset Sizes

normal/shuffle/blank/text_only n=2195; relcomp n=666; facingcomp n=103; hflip_flip n=245; hflip_invariant n=421. All verified against frozen eligible-IDs manifests.

5. Archived Artifacts

All raw predictions, analysis JSONs, checkpoints, training logs, figures, and audit outputs are archived under results/seed_campaign/ in the repository (freeze commit 95b66bc on research/spatial-grounding-audit).